

Prototyping Educational Innovation



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What do you think of this idea that we're possibly going to prototype in a few weeks during a week-long 8th grade teams project? (This is, as opposed to something very Asana-like.)

I think this is genuinely interesting — not because it's “AI in education” (which is already overcrowded), but because it attacks a very specific coordination problem that most ed-tech systems completely miss: the tension between *adaptive scaffolding* and *shared physical collaboration*.

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A few things strike me as especially strong:

The paper-loop idea is the real innovation

Most people reading this will initially think “AI tutor” or “workflow automation,” but I actually think the core invention is the *closed paper-feedback loop*. The sheets are not just outputs; they are the state representation of the system. That's unusual and smart.

You're effectively treating paper as:

- the collaborative UI,
- the persistent local state,
- the social object,
- and the AI sensing surface.

That's much more novel than “AI generates worksheets.”

There's also a deep HCI insight here: paper naturally supports ambiguity, annotation, side-conversations, pointing, partial completion, and shared attention in ways screens are surprisingly bad at for middle-school groups.

You're implicitly exploiting decades of CSCW/HCI findings without sounding academic about it.

The “per-team conversational memory” idea is stronger than you may realize

This part:

“No database captures this — it lives in the conversation.”

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...is provocative in a good way.

Most project-management systems reduce state to explicit fields:

- task completed,
- due date,
- assignment,
- status.

But actual team trajectories are narrative and contextual:

- “they got demoralized,”
- “they keep reinterpreting the goal,”
- “one kid dominates discussion,”
- “they're creatively ahead but organizationally behind.”

LLM conversational state is oddly well-suited to exactly this fuzzy longitudinal memory problem.

Ironically, this may be one of the few domains where “soft memory” is more valuable than rigid databases.

The “Wizard” framing is excellent

Very smart to foreground the Wizard-of-Oz structure instead of pretending the AI is autonomous.

That does three important things:

1. avoids overclaiming,
2. makes the prototype feasible,
3. keeps pedagogy human-centered.

And practically: the Wizard becomes a *curriculum steering layer*.

I suspect that in practice the OzNotes may become the most important part of the system. They are basically compressed pedagogical interventions.

It's anti-Asana in exactly the right way

You mentioned “as opposed to something very Asana-like.”

Yes. And that distinction matters.

Asana/Jira/Trello assume:

- tasks are known,
- workflow is stable,
- progress is legible,
- deviations are errors.

But classroom projects are exploratory social systems. The “messiness” is the point.

Your system treats divergence as signal rather than failure.

That's much closer to how good teachers actually operate.

I particularly like that the students never see the AI

That's subtle but important.

The AI is functioning as:

- backstage orchestration,
- adaptive scaffolding,
- teacher augmentation.

Not as:

- authority,
- conversational companion,
- answer machine.

That avoids a huge number of pedagogical and ethical problems immediately.

It also keeps the *social center of gravity* inside the student team rather than between student and machine.

That's rare in AI-ed.

A few concerns / challenges I'd watch carefully:

1. OCR/handwriting robustness may dominate the prototype

Not conceptually — operationally.

Middle-school handwriting plus sketches plus partial checkmarks plus arrows plus jokes in margins could become chaotic fast.

You may want the first prototype to:

- heavily structure the writable regions,
- prefer checkboxes/circles/short text,
- and intentionally ignore most freeform writing initially.

Otherwise the AI ingestion pipeline could become the bottleneck.

2. Conversation drift / hallucinated team narratives

A long-running conversational memory can become *too interpretive*.

The AI may start inventing stable traits:

- “Team Gamma is weak at planning”
- “Alpha struggles with deadlines”

...based on sparse evidence.

You'll probably need:

- periodic grounding summaries,
- explicit state extraction,
- and maybe “evidence tagging.”

Otherwise the conversational memory may become self-reinforcing folklore.

3. The nightly Wizard loop may be more cognitively expensive than expected

The system's success depends heavily on whether the Wizard summaries are:

- concise,
- trustworthy,
- and actionable.

If the Wizard has to read five quasi-psychological narratives nightly, fatigue will kill the workflow.

You probably want:

- highly compressed summaries,
- exception-focused reporting,
- and confidence indicators.

4. The Gantt chart may matter less than the narrative framing

Middle-schoolers often don't internalize timelines abstractly.

But they *do* respond to:

- “you're almost ready for X,”
- “today is about making the idea concrete,”
- “your team solved the hard part yesterday.”

I suspect the “where we are / where we're going” paragraph may become disproportionately important.

There's also a deeper philosophical angle here that I think is worth emphasizing eventually:

Most educational AI systems try to optimize *instruction*.

TedGy is trying to optimize *coordination and continuity*.

That's a very different thing.

It's closer to:

- cognitive apprenticeship,
- distributed cognition,
- scaffolding,
- project memory,
- and collaborative regulation.

The AI is acting less like a tutor and more like a lightweight external executive function for the team.

That's potentially very important.

And honestly, a week-long 8th grade prototype is exactly the right scale. Long enough for team divergence and memory effects to emerge, short enough that operational problems won't bury you.